

JOHN M. REKOSKE, PH.D.

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PROFESSIONAL SUMMARY

Computational geophysicist specializing in physics-aware machine learning and reduced-order modeling (ROM) for seismological applications. Demonstrated expertise designing surrogate and ROM frameworks that accelerate high-fidelity physics simulations (elastic wave propagation, seismic hazard analysis) by orders of magnitude, enabling real-time inference on large-scale problems. Strong scientific record with 15 peer-reviewed publications and substantial experience with HPC environments, deep learning, and scientific software development in Python, C++, and Fortran. Proven ability to bridge AI/ML research and domain science communities, with experience collaborating across geophysics, engineering, and computational science.

EDUCATION

University of California, San Diego, La Jolla, CA
Colorado School of Mines, Golden, CO

Ph.D. Earth Sciences, 2026
B.S. Geophysical Engineering, 2020

WORK EXPERIENCE

Caltech Seismological Laboratory Director's Postdoctoral Research Associate in Geophysics.

California Institute of Technology; Pasadena, CA; 2026-Present

Leads research on next-generation earthquake early warning algorithms and methodology using surrogate models, large heterogeneous datasets, and physics-aware machine learning to support real-time, operational warning and monitoring for the safety of critical infrastructure.

National Science Foundation Graduate Research Fellow in Geophysics.

Scripps Institution of Oceanography; La Jolla, CA; 2021-2026

Conducted geophysical research on earthquake ground motions using machine learning and novel reduced-order models to accelerate numerical elastic wave propagation simulations for earthquake early warning, seismic hazard analysis, and to better understand earthquake source and rupture processes. Published 3 first-author research articles, co-authored 4 research articles, and published 3 seismic datasets.

Physical Scientist in Earthquake Hazards Program

United States Geological Survey; Golden, CO; 2018-2021

Core developer of official USGS software. Developed a research prototype into production-grade operational software deployed for national seismic monitoring and seismic data workflows including ML-based inverse methods for ground motion parameter estimation. Compiled datasets of earthquake ground motion data and analyzed ground motion models to improve understanding of shaking and structural dynamics.

SKILLS

- Scientific Computing & HPC: Python (NumPy, SciPy, Pandas, Matplotlib, ObsPy), C/C++, Fortran, R, SQL; MPI/OpenMP parallel workflows; TACC Frontera, LRZ SuperMUC-NG; SDSC Expanse; HDF5, Conda, Git
- AI/ML Frameworks & Methods: PyTorch, TensorFlow/Keras, Scikit-learn; physics-informed neural networks (PINNs); reduced-order modeling (ROM/POD); surrogate modeling; convolutional neural networks (CNNs); inverse problem methods; uncertainty quantification; V&V of ML models for physical consistency; feature engineering for time-series/waveform data
- Physics Simulation Platforms: elastic wave propagation; finite element modeling; discontinuous Galerkin method; ground motion simulation; seismic hazard analysis, SeisSol, SPECFEM3D, SPECFEM3D_GLOBE

AWARDS

- Caltech Seismological Laboratory Director's Fellowship
- National Science Foundation Graduate Research Fellowship
- Scripps Institution of Oceanography Institutional Fellowship
- Seismological Society of America Travel Grant Award
- Colorado School of Mines Cecil H. Green Gold Medal Award

SERVICE

- Peer reviewer for *Bulletin of the Seismological Society of America*, *Seismological Research Letters*, *Journal of Geophysical Research: Solid Earth*, and *Earthquake Spectra*.
- Session convener at the American Geophysical Union 2023 Annual Meeting.
- Quakeworx developer instructor at the Statewide California Earthquake Center 2025 Annual Meeting.

PUBLICATIONS (H-INDEX=12)

1. **Rekoske, J. M.**, Callaghan, S., Milner, K., May, D., and Gabriel, A.-A. (2026). Efficient Physics-Informed Ground-Motion Simulations with Reduced-Order Models: CyberShake Implications and High-Resolution Site Terms for Southern San Andreas Fault Earthquakes, *Bulletin of the Seismological Society of America*.
2. **Rekoske, J. M.**, May, D. A., and Gabriel, A.-A. (2025). Reduced-order modeling for complex 3D seismic wave propagation. *Geophysical Journal International* 241, no. 1, 526-548.
3. Thompson, E., Hearne, M., Aagaard, B. T., Rekoske, J. M., Worden, C. B., Moschetti, M. P., ... and Kottke, A. R. (2024). Automated, Near Real-Time Ground-Motion Processing at the US Geological Survey. *Seismological Research Letters*.
4. **Rekoske, J. M.**, Gabriel, A.-A., & May, D. A. (2023). Instantaneous Physics-Based Ground Motion Maps Using Reduced-Order Modeling. *Journal of Geophysical Research: Solid Earth* 128, no. 8, e2023JB026975.
5. Gabriel, A.-A., T. Ulrich, M. Marchandon, J. Biemiller, and **Rekoske, J. M.** (2023). 3D Dynamic Rupture Modeling of the 6 February 2023, Kahramanmaraş, Turkey M w 7.8 and 7.7 Earthquake Doublet Using Early Observations. *The Seismic Record* 3 no. 4, 342–356.
6. Jia, Z., Z. Jin, M. Marchandon, T. Ulrich, A.-A. Gabriel, W. Fan, P. Shearer, X. Zou, **Rekoske, J. M.**, F. Bulut, et al. (2023). The complex dynamics of the 2023 Kahramanmaraş, Turkey, Mw 7.8-7.7 earthquake doublet, *Science* 381, no. 6661, 985–990.
7. Nayak, A., V. Rodríguez Tribaldos, J. Ajo-Franklin, B. Miranda, C.-C. Chien, R. Mellors, M. Robertson, M. Brandin, **Rekoske, J. M.**, and T. Wood (2023). Nodal and Broadband Seismometer Complement to the Imperial Valley Dark Fiber DAS Array, *Seismological Research Letters* 94, no. 6, 2852–2867.
8. **Rekoske, J. M.**, M. P. Moschetti, and E. M. Thompson (2022). Basin and site effects in the US Pacific Northwest estimated from small-magnitude earthquakes, *Bulletin of the Seismological Society of America* 112, no. 1, 438–456.
9. Kleckner, J. K., K. B. Withers, E. M. Thompson, **Rekoske, J. M.**, E. Wolin, and M. P. Moschetti (2022). Automated detection of clipping in broadband earthquake records, *Seismological Research Letters* 93, no. 2A, 880–896.
10. Moschetti, M. P., D. Churchwell, E. M. Thompson, **Rekoske, J. M.**, E. Wolin, and O. S. Boyd (2021). Seismic wave propagation and basin amplification in the Wasatch Front, Utah, *Seismological Research Letters* 92, no. 6, 3626–3641.
11. **Rekoske, J. M.**, E. M. Thompson, M. P. Moschetti, M. G. Hearne, B. T. Aagaard, and G. A. Parker (2020). The 2019 Ridgecrest, California, earthquake sequence ground motions: Processed records and derived intensity metrics, *Seismological Research Letters* 91, no. 4, 2010–2023.
12. Moschetti, M. P., E. M. Thompson, **Rekoske, J. M.**, M. G. Hearne, P. M. Powers, D. E. McNamara, and C. Tape (2020). Ground-Motion amplification in Cook Inlet Region, Alaska, from intermediate-depth earthquakes, including the 2018 Mw 7.1 Anchorage earthquake, *Seismological Research Letters* 91, no. 1, 142–152.
13. Parker, G. A., A. S. Baltay, **Rekoske, J. M.**, and E. M. Thompson (2020). Repeatable source, path, and site effects from the 2019 M 7.1 Ridgecrest earthquake sequence, *Bulletin of the Seismological Society of America* 110, no. 4, 1530–1548.
14. McNamara, D. E., Wolin, E., Powers, P. M., Shumway, A. M., Moschetti, M. P., **Rekoske, J. M.**, ... & Petersen, M. D. (2020). Evaluation of Ground-Motion Models for US Geological Survey Seismic Hazard

Models: 2018 Anchorage, Alaska, Mw 7.1 Subduction Zone Earthquake Sequence, *Seismological Research Letters* 91, no. 1, 183-194.

15. McNamara, D. E., Wolin, E., Powers, P. M., Shumway, A. M., Moschetti, M. P., **Rekoske, J. M.**, ... & Petersen, M. D. (2020). Evaluation of ground-motion models for US Geological Survey seismic hazard forecasts: Hawaii tectonic earthquakes and volcanic eruptions. *Bulletin of the Seismological Society of America* 110, no. 2, 666-688.

PUBLISHED GEOPHYSICAL DATASETS

1. **Rekoske, J. M.**, May, D. A., & Gabriel, A.-A. (2024). Dataset for "Reduced-order modeling for complex 3D seismic wave propagation". Zenodo.
2. **Rekoske, J. M.**, Gabriel, A.-A., & May, D. A. (2023). jrekoske/Instantaneous-PhysicsBased-GroundMotionMaps: Publication release (v1.0.0). Zenodo.
3. **Rekoske, J. M.**, Moschetti, M. P., & Thompson, E. M. (2019). Database of ground motions from tectonic and volcanic events, Hawaii, 2018. Center for Engineering Strong Motion Data (CESMD).
4. **Rekoske, J. M.**, Thompson, E. M., Moschetti, M. P., Hearne, M. G., Aagaard, B. T., & Parker, G. A. (2019). Ground motions from the 2019 Ridgecrest, California, earthquake sequence. Center for Engineering Strong Motion Data (CESMD).
5. **Rekoske, J. M.**, Moschetti, Morgan P., Thompson, Eric M and Frankel, Arthur D. (2021). Database of horizontal-component Fourier amplitude spectra of acceleration ground motions from Pacific Northwest earthquakes, 1999-2020. Center for Engineering Strong Motion Data (CESMD).
6. Churchwell, David, Moschetti, Morgan P., Thompson, Eric M., **Rekoske, J. M.**, Wolin, Emily, and Boyd, Oliver (2021). Database of horizontal-component Fourier amplitude spectra of acceleration ground motions from Utah earthquakes, 2010-2020. Center for Engineering Strong Motion Data (CESMD).
7. **Rekoske, J. M.**, Morgan P. Moschetti, Eric M. Thompson, Mike G. Hearne, Carl Tape (2019). USGS database of ground motions from in-slab earthquakes near Anchorage, Alaska, 2008-2019. Center for Engineering Strong Motion Data (CESMD).

INVITED TALKS AND SEMINARS

1. **Rekoske, J. M.**, L. Kuratle, D. Kilb, & A.-A. Gabriel. Improving Scientific Communication using Earthquake Simulations: Insights from a Public Survey. American Geophysical Union Annual Meeting. December 2025. New Orleans, LA.
2. **Rekoske, J. M.**, A.-A. Gabriel, & D. A. May. Rapid physics-informed wavefields using high-performance computing and reduced-order models. United States Geological Survey, Earthquake Science Center Seminar. September 2024. Moffett Field, CA.
3. **Rekoske, J. M.**, A.-A. Gabriel, & D. A. May. Rapid physics-informed wavefields using high-performance computing and reduced-order models. Megathrust Modeling Framework, All-hands Science Discussion. October 2024.